

KAIST
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[MAS456] Statistical Methods with Computer
Homework 4

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Chapter 8 - Tree-Based Methods

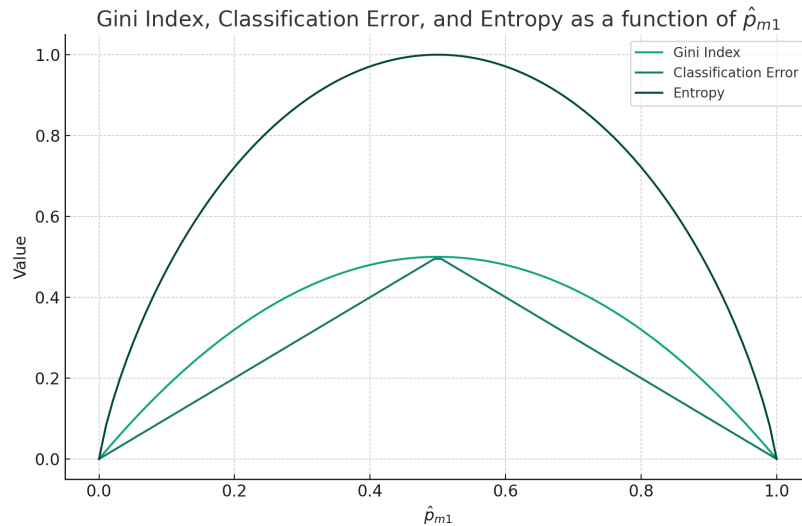
Problem #2

$$f(x) = \sum_{b=1}^B \lambda \hat{f}^b(x) \quad (1)$$

In this equation, $\hat{f}^b(x)$ denotes the function of the b th tree, with each tree having just one split. Since trees of depth 1 are considered only a single variable, the total function for x is formed by aggregating the outputs from each individual tree, indicating that the model is additive. If the trees were of depth 2, they would be capable of capturing interactions between two variables.

Problem #3

Plot:



Method:

The Python code used to generate the plot is as follows:

```
import numpy as np
import matplotlib.pyplot as plt

# Define p_m1, the probability of class 1
p_m1 = np.linspace(0, 1, 100)

# Calculate the corresponding probability of class 2
p_m2 = 1 - p_m1

# Calculate the Gini index, classification error, and entropy
gini_index = 1 - (p_m1**2 + p_m2**2)
```

```

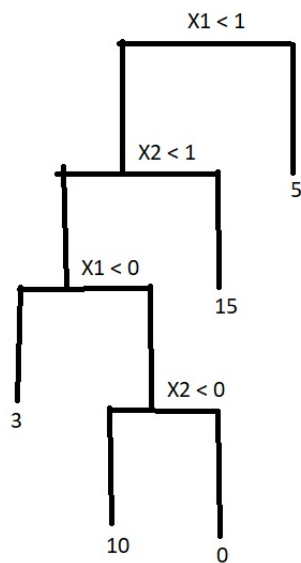
classification_error = 1 - np.maximum(p_m1, p_m2)
entropy = - (p_m1 * np.log2(p_m1 + 1e-9) + p_m2 * np.log2(p_m2 + 1e-9))

# Plotting
plt.figure(figsize=(10, 6))
plt.plot(p_m1, gini_index, label='Gini Index')
plt.plot(p_m1, classification_error, label='Classification Error')
plt.plot(p_m1, entropy, label='Entropy')
plt.title('Gini Index, Classification Error, and Entropy as a function of  $\hat{p}_{m1}$ ')
plt.xlabel(' $\hat{p}_{m1}$ ')
plt.ylabel('Value')
plt.legend()
plt.grid(True)
plt.show()

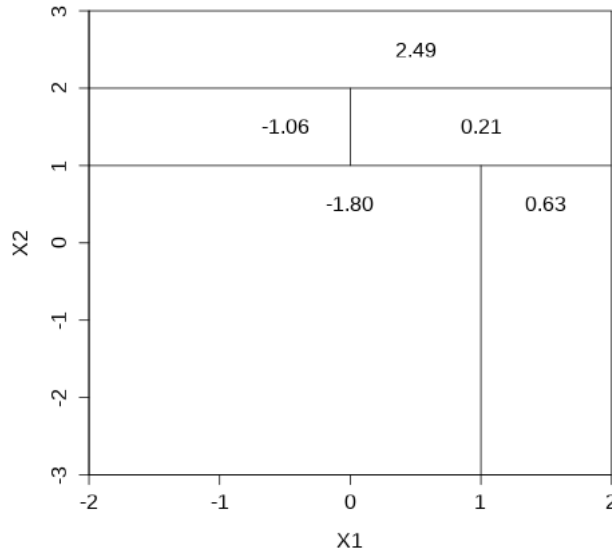
```

Problem #4

(a) **Plot:**



(b) **Plot:**



Problem #5

Given the probabilities of a class being red from ten bootstrapped samples:

0.1, 0.15, 0.2, 0.2, 0.55, 0.6, 0.6, 0.65, 0.7, 0.75,

we consider two approaches for final classification:

1. **Majority Vote Approach:** Classify as red if the majority of probabilities indicate red (> 0.5).

In this case, there are 6 out of 10 probabilities greater than 0.5, thus the classification is red.

$$\text{Number of estimates} > 0.5 = 6 > \frac{10}{2} \implies \text{Classify as Red}$$

2. **Average Probability Approach:** Classify as red if the average probability is greater than 0.5. The average probability is:

$$\frac{0.1 + 0.15 + 0.2 + 0.2 + 0.55 + 0.6 + 0.6 + 0.65 + 0.7 + 0.75}{10} = 0.45$$

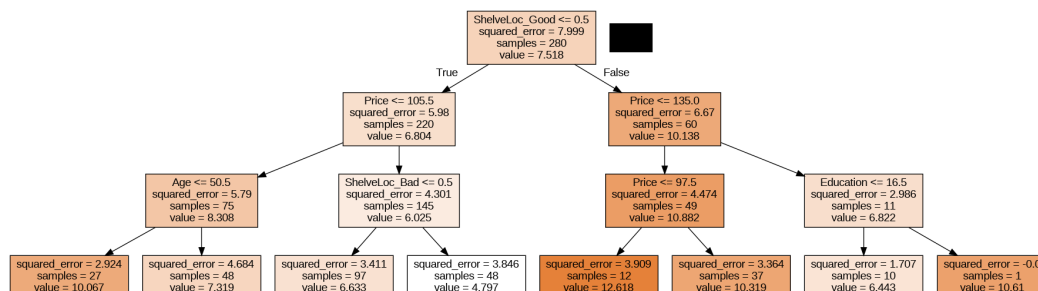
Since 0.45 is less than 0.5, the classification is not red (So, green).

$$\text{Average Probability} = 0.45 < 0.5 \implies \text{Classify as Not Red}$$

Problem #8

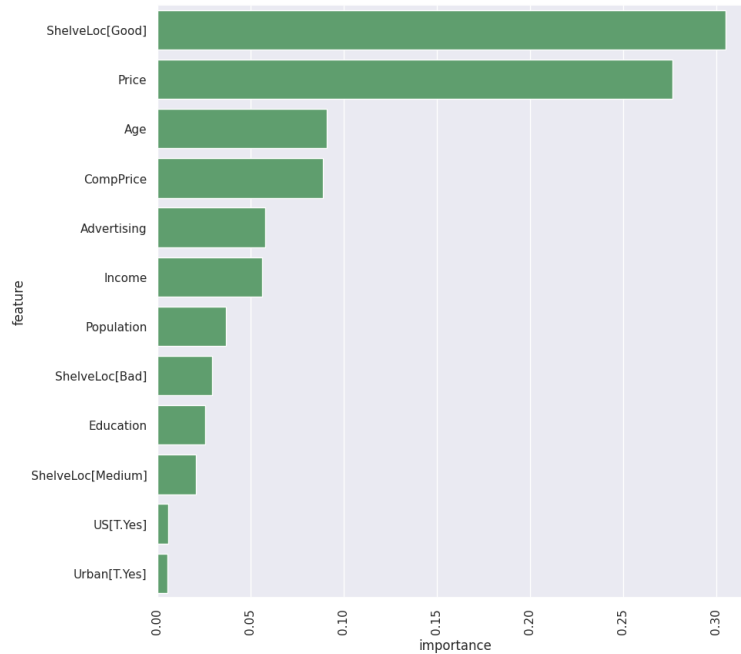
(a) Done

(b) The Tree:

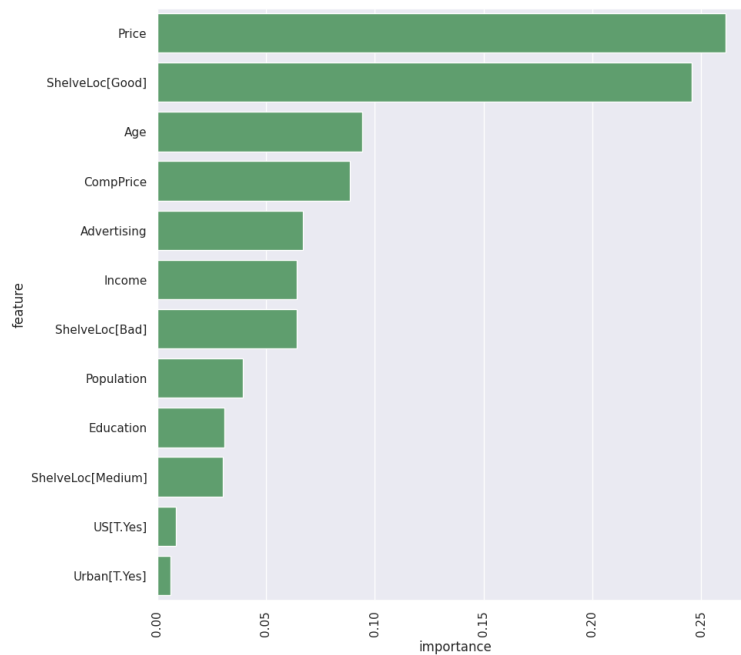


The Test MSE comes out to be 5.09.

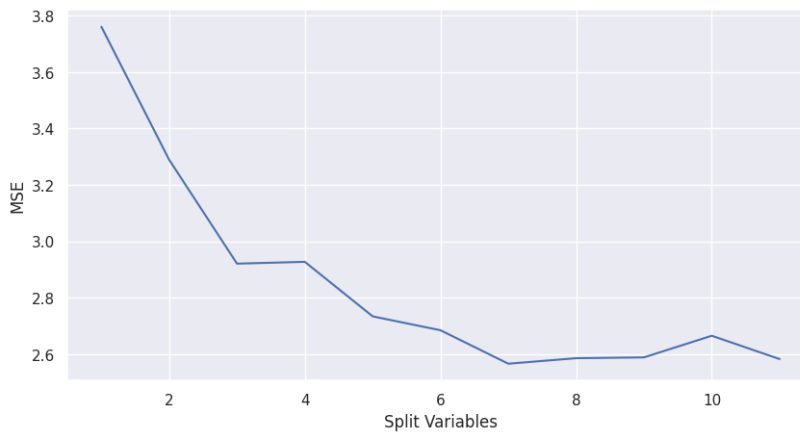
- (c) The Test MSE after pruning is 4.661, which is an improvement over 5.09.
- (d) The Test MSE following the bagging approach is 2.615, a huge improvement over the previous scenario. On top of that, the bagging model suggests that ShelfLoc[Good] and Price are the most important features of this dataset in determining Sales revenue.



- (e) Random forest with 7 predictors does an even better job. The test MSE is 2.544. Not a huge improvement but there are some improvements at least.



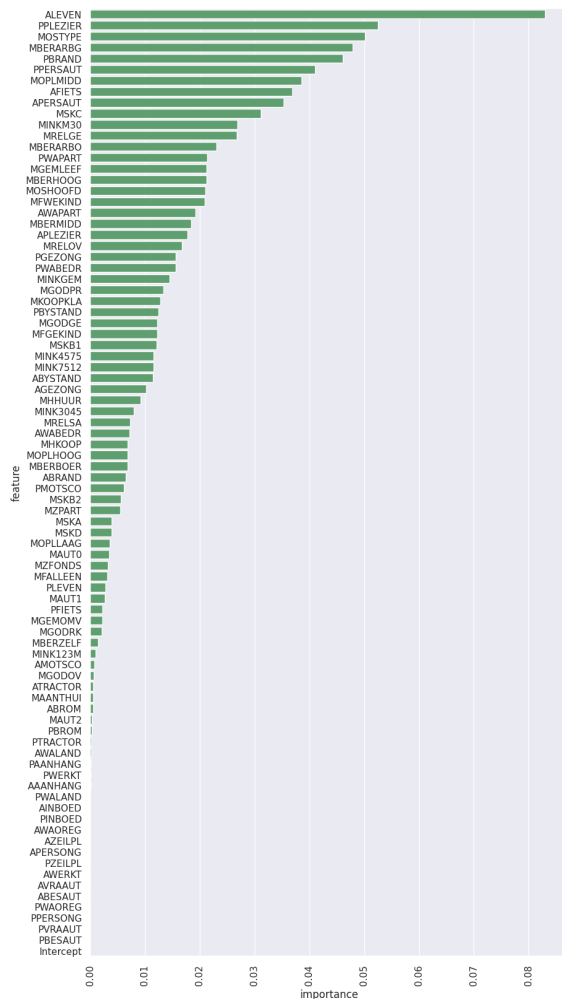
For feature importance, it swaps the top 2 features among themselves (Price and ShelfLoc[Good]) and no new feature is marked as important.



As we can see from the chart, as the value of m increases, we see a roughly steady decline in MSE.

Problem #11

- (a) Done
- (b) PPLEIZER, ALEVAN, PBRAND, MOSTYPE, MBERARBG turns out to be the 5 most significant indicators to predict if a customer will buy caravan insurance or not. There are other values that gradually follow these indicators. All of them can be seen from the chart:



(c) **Boosting Model:**

Confusion Matrix:

		Predicted	
		Positive	Negative
Actual	Positive	4403	122
	Negative	264	33

Positive Predictive Value (PPV) = 0.2129

KNN:

KNN produces the best result when K=2. We get a confusion matrix that is like:

		Predicted	
		Positive	Negative
Actual	Positive	4507	18
	Negative	293	4

Positive Predictive Value (PPV) = 0.18182

So, the boosting model does a better job than KNN in this scenario although both of their PPV is kinda close. One reason behind KNN not performing as well can be the high dimensionality of the dataset (80+ predictors).

Chapter 11 - Survival Analysis and Censored Data

Problem #2

Participant 1:

- (a) No, the replacement time is not censored.
- (b) No, c_1 is not known.
- (c) Yes, t_1 is 1.2 years.
- (d) Yes, $y_1 = \min(c_1, t_1) = 1.2$ years.
- (e) Yes, $\delta_1 = 1$ (event occurred).

Participant 2:

- (a) Yes, the replacement time is censored.
- (b) Yes, c_2 is 2 years.
- (c) No, t_2 is not known.
- (d) Yes, $y_2 = \min(c_2, t_2) = 2$ years.
- (e) Yes, $\delta_2 = 0$ (censored).

Participant 3:

- (a) Yes, the replacement time is censored.
- (b) Yes, c_3 is 1.5 years.
- (c) No, t_3 is now known.
- (d) Yes, $y_3 = \min(c_3, t_3) = 1.5$ years.
- (e) Yes, $\delta_3 = 0$ (censored).

Participant 4:

- (a) No, the replacement time is not censored.
- (b) No, c_4 is not known.
- (c) Yes, t_4 is 0.2 years.
- (d) Yes, $y_4 = \min(c_4, t_4) = 0.2$ years.
- (e) Yes, $\delta_4 = 1$ (event occurred).

Problem #3

- Number of distinct event times (K): $K = 2$ (events at 0.2 and 1.2 years).
- Number of events ($d_1, d_2, \dots$):
 - At 0.2 years, 1 phone replacement ($d_1 = 1$).
 - At 1.2 years, another phone replacement ($d_2 = 1$).
- Number at risk ($r_1, r_2, \dots$):
 - At start (0 years), all 4 participants are at risk ($r_0 = 4$).
 - Just before 0.2 years, all 4 are still at risk ($r_1 = 4$).
 - Just before 1.2 years, 3 participants are at risk ($r_2 = 3$).
- Survival probability ($q_1, q_2, \dots$):
 - At 0.2 years, 1 person dies ($q_1 = 1$).
 - At 1.2 years, 1 person dies ($q_2 = 1$).

Problem #4

(a) Given data:

Observation (Y)	Censoring Indicator(δ)
26.5	1
37.2	1
57.3	1
90.8	0
20.2	0
89.8	0

Ordered by observation time:

20.2	0
26.5	1
37.2	1
57.3	1
89.8	0
90.8	0

Survival probabilities calculated at each time point:

Time	Survival Probability
20.2	(Censored)
26.5	$\frac{4}{5}$
37.2	$\frac{4}{5} \times \frac{3}{4}$
57.3	$\frac{4}{5} \times \frac{3}{4} \times \frac{2}{3}$
89.8	(Censored)
90.8	(Censored)

Thus, the estimated survival probability past 50 days is:

$$S(50) = \frac{4}{5} \times \frac{3}{4} = 0.6 \quad (\text{or } 60\%)$$

(b)

$$\hat{S}(t) = \begin{cases} 1 & \text{if } t < 26.5 \\ 0.8 & \text{if } 26.5 \leq t < 37.2 \\ 0.6 & \text{if } 37.2 \leq t < 57.3 \\ 0.4 & \text{if } 57.3 \leq t \end{cases}$$

Problem #7

(a) The hypergeometric distribution is applicable in scenarios where we are dealing with a finite population and sampling is done without replacement. In our case, we are interested in the number of failures (q_{1k}) when we randomly select r_{1k} observations from a risk set of r_k observations, which includes q_k failures. This scenario aligns with the principles of the hypergeometric distribution.

In terms of the hypergeometric distribution parameters, we have:

- N (the total population size) corresponds to r_k .
- K (the total number of successes in the population) corresponds to q_k .
- n (the number of draws) corresponds to r_{1k} .
- k (the number of observed successes in the sample) corresponds to q_{1k} .

Therefore, q_{1k} follows a hypergeometric distribution with parameters r_k (population size), q_k (number of successes in the population), and r_{1k} (number of draws).

(b) Given the properties of the hypergeometric distribution, the mean and variance of q_{1k} are calculated as follows:

- **Mean Calculation:** The mean of a hypergeometric distribution is given by $n \frac{K}{N}$. For our problem, this is represented as:

$$\text{Mean} = r_{1k} \frac{q_k}{r_k}. \quad (2)$$

- **Variance Calculation:** The variance of a hypergeometric distribution is $n \frac{K}{N} \left(1 - \frac{K}{N}\right) \frac{N-n}{N-1}$. In our context, the variance of q_{1k} is:

$$\text{Variance} = r_{1k} \frac{q_k}{r_k} \frac{r_k - q_k}{r_k} \frac{r_k - r_{1k}}{r_k - 1}. \quad (3)$$

These expressions align with the general properties of the hypergeometric distribution and are tailored to the specific parameters of our distribution, r_{1k} , q_k , and r_k .

Problem #8

Given that $F(t) = 1 - S(t)$, where $F(t)$ is the cumulative distribution function (CDF) and $S(t)$ is the survival function. The relationship between the probability density function (pdf) $f(t)$ and the CDF $F(t)$ in a continuous setting is given by:

$$f(t) = \frac{dF(t)}{dt} \quad (4)$$

Since $F(t) = 1 - S(t)$, the derivative of $F(t)$ with respect to t is the negative derivative of $S(t)$ with respect to t . Thus,

$$f(t) = -\frac{dS(t)}{dt} \quad (5)$$

Now, considering the hazard function $h(t)$, which is defined as $h(t) = \frac{f(t)}{S(t)}$, we substitute the expression for $f(t)$ from equation (2):

$$h(t) = -\frac{\frac{dS(t)}{dt}}{S(t)} = -\frac{d}{dt} \log S(t) \quad (6)$$

Integrating both sides with respect to t gives us:

$$\int h(t) dt = -\int \frac{d}{dt} \log S(t) dt \quad (7)$$

Integrating the left-hand side from 0 to t and the right-hand side by parts, we obtain:

$$\int_0^t h(u) du = -\log S(t) \quad (8)$$

Exponentiating both sides to eliminate the logarithm, we arrive at the second identity:

$$S(t) = \exp \left(-\int_0^t h(u) du \right) \quad (9)$$

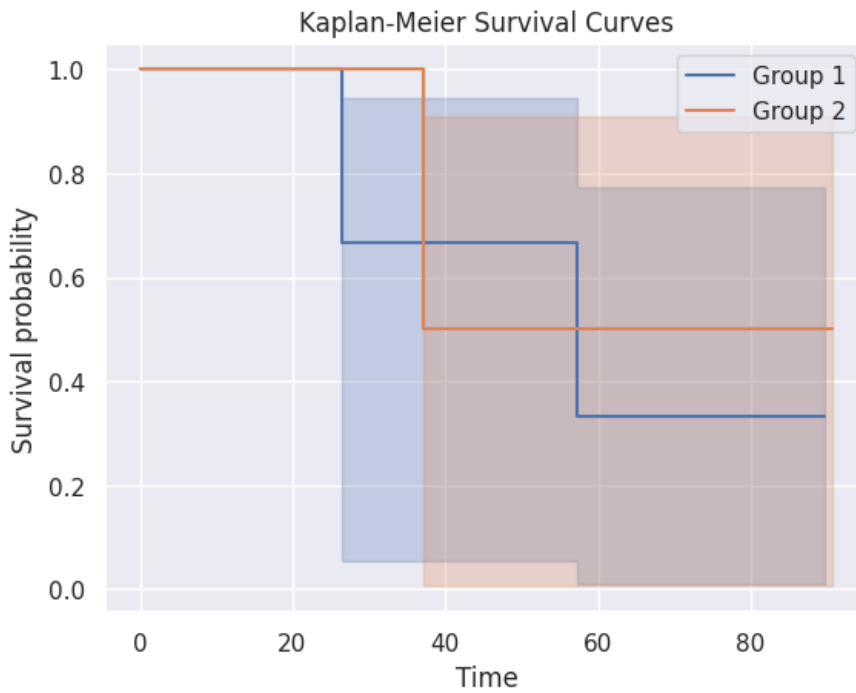
These steps demonstrate the required relationships between $f(t)$, $S(t)$, and $h(t)$.

Problem #11

(a) Kaplan-Meier Survival Curves

Kaplan-Meier survival curves were plotted for two groups based on the covariate X . Group 1 includes observations with $X < 2$ and Group 2 includes those with $X \geq 2$. The survival curves are visual representations of the survival probability over time for each group. There isn't any significant difference

between the curves.



(b) Cox Proportional Hazards Model

A Cox proportional hazards model was fitted using the group indicator as a covariate. The model output is as follows:

coef	exp(coef)	se(coef)	z	p	-log2(p)
-0.34	0.71	1.24	-0.28	0.78	0.35

The estimated coefficient for Group 2 is -0.34 , with an exponentiated coefficient (hazard ratio) of 0.71 . However, the wide confidence interval and high p-value indicate that the difference in hazard rates between Group 1 and Group 2 is not statistically significant. This high p-value indicates that there is no strong evidence to reject the null hypothesis; thus, we do not have sufficient evidence to claim that the true coefficient is different from zero. The coefficient's standard error is also quite large relative to the coefficient itself, which further weakens the evidence against the null hypothesis.

(c) Log-Rank Test

The log-rank test was conducted to compare the survival distributions between the two groups. The test yielded a p-value of approximately 0.782 , suggesting no significant difference between the survival curves of the two groups.

Interpretation

Both the Cox model and the log-rank test suggest that there is no statistically significant difference in survival between the two groups. The p-value for the log-rank test statistic is approximately 0.782 ,

and the p-value for the score statistic from the Cox model is also 0.78. Both p-values are well above the conventional alpha level of 0.05, indicating that neither test found a statistically significant difference between the survival curves of the two groups.

In survival analysis, when a single binary covariate is involved, the log-rank test statistic is mathematically equivalent to the score statistic of the Cox model. Therefore, it is expected that both would yield similar p-values, which is indeed what we observe in our output.

The consistency of p-values between the Cox model and the log-rank test reinforces the conclusion that there is no statistically significant evidence to suggest a difference in survival probabilities between the two groups based on the provided dataset.

The data does not provide evidence to support a non-zero coefficient, meaning that the covariate X does not significantly affect survival probabilities in the given dataset.